Afshaal Zubair

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Education

University of Texas at Austin *Aug 2024 to Dec 2027*
Bachelor of Science in Computer Science

- **Coursework:** Data Structures, Discrete Math, Computer Architecture, Operating Systems, Algorithms and Complexity, Programming for Performance, Game Programming, Ethical Hacking, Pitch to Prototype

Skills

Languages: C/C++, Java, Python, SQL, TypeScript/JavaScript, HTML, CSS

Frameworks/Libraries: Node.js, React.js, PyTorch, Pygame, Matplotlib, NumPy, Next.js

Developer Tools: Linux Toolchain, Git, Bash, PowerShell, Wireshark, VirtualBox, Nmap, JSON, VSCode, Vite

Other Skills: Microsoft Office 365, Excel, Adobe Premiere Pro, Spline 3D

Projects

Ride Brief *Github Repo*

- Modeled the NHTSA FARS 2023 fatal-crash files in DuckDB and wrote SQL marts with multi-table joins, CTEs, and window functions for 6,367 motorcyclists killed in 6,275 crashes, matching the published helmet split of 4,018 used, 2,124 not used, and 225 unknown.
- Built an Excel briefing a non-technical reader can audit, including a data dictionary, one row per death, and COUNTIFS state and national totals checked against the SQL warehouse.
- Shipped a dashboard of the same metrics, left police-reported alcohol missing for 3,000 deaths instead of imputing it, and ranked states by death counts because the file has no miles-ridden exposure.

Foundation Plate *Github Repo*

- Formulated a mixed-integer menu model (PuLP/CBC) that fills 28 meals from 27 USDA Foundation Foods, holding each day within 10% of a calorie target and at least 95% of a protein target under a weekly grocery budget.
- Built a Next.js and FastAPI planner that sets targets from body stats and training days, supports food swaps and price edits, and shows remaining gaps against adult intakes for fiber, iron, potassium, calcium, and magnesium.
- Aggregated 4.06 million OpenPowerlifting meet rows to one best total per lifter (420,954 lifters) and compared the user's body weight with IPF class medians for squat, bench, deadlift, and total.

Conway's Conundrum *Github Repo*

- A twist on the famous cellular automaton "Conway's Game of Life", which allows the user to experiment with a wide variety of custom parameters to create novel simulations.
- Built in Python using the Pygame library.

Experience

Founder and Secretary *Houston, TX*
Snowdrop United Corporation *Jul 2022 to present*

- Kick-started a fundraising campaign and nonprofit organization to help those with food insecurity gain access to meals within the greater Houston area.
- Fund-raised over $2,750 in donations. Successfully prepared, packaged, and distributed over four carloads of goods accumulating to over 1500 meal and hygiene packages across 9 Hope Events.
- Handled corporation logistics, tax filing, banking, social media/website outlook, and PayPal fundraising.

Deep Learning Student *Austin, TX*
Directed Reading Program *Aug 2024 to Dec 2024*

- Orchestrated project based deep learning solutions, followed mentored research in meetings one hour per week.
- Explored a wide variety of machine learning/AI concepts including neural networks, CNNs, transformers, and diffusion models. Strengthened ability in developing deep learning models through the PyTorch framework.